

Shaunak Sanjay More

AI / Machine Learning Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer experienced in building deep learning models, AI systems, and data infrastructure. Strong background in computer vision, generative AI systems, and large-scale AI pipelines using PyTorch, LangGraph, and vector databases. Winner of two national-level project competitions for galaxy morphology classification and photometric redshift estimation.

TECHNICAL SKILLS

Programming Languages: C++, Python, JavaScript

ML Frameworks and Techniques: PyTorch, TensorFlow, Scikit-learn, Deep Learning, CNNs, Transformers, Computer Vision, Representation Learning

AI Systems: LangGraph, RAG pipelines, Vector Databases, LLM Agents

Infrastructure: Docker, Kubernetes, AWS

Databases: SQL, MongoDB, Pinecone, Neo4j

Web and Backend frameworks: FastAPI, Node.js, React, WebSockets

Others: DSA, DBMS

PROFESSIONAL EXPERIENCE

IT and Cyber-Security Intern | Dec 2023 - Jan 2024

Five Holdings | Dubai

- Improved asset lifecycle management workflows using enterprise UEM/MDM tools.
- Standardized device provisioning pipelines reducing manual onboarding overhead.

Technologies Used: ManageEngine UEM, Mobile MDM

PROJECTS

Galaxy Morphology Classification & Photometric Redshift Estimation

Winner – IEEE Technovation & MET Enthusia Project competitions | [GitHub](#)

- Developed CNN-based model for **galaxy morphology classification** using multi-band astronomical images
- Trained on **40k galaxy images from the Sloan Digital Sky Survey (SDSS)** across 5 photometric bands (**g, r, i, z, y**).
- Trained models using PyTorch with **GPU acceleration** and **K-Fold cross-validation**.
- Achieved **90.19% validation accuracy** on galaxy morphology classification.
- Built hybrid model combining **photometric features and deep image embeddings** for photometric redshift estimation.
- Achieved an **MSE of 0.001197** and **MAE of 0.085** for redshift estimation.

Mini-Genie — CNN-Transformer World Model for Video Prediction

[GitHub](#)

- Developed a **CNN-Transformer architecture in PyTorch** for action-conditioned video frame prediction using frames generated from **OpenAI Gymnasium reinforcement learning environments**.
- CNN encoder extracts spatial features while Transformer models **temporal dependencies across frames**.
- Implemented **autoregressive prediction pipeline** generating future frames conditioned on action inputs.
- Optimized training pipeline with batched augmentations and checkpointing, reducing training time by **20%**.
- Achieved an **MSE of 0.000447**, **MAE of 0.001996** and a **Peak Signal-to-Noise Ratio of 33.50 dB**

ShoreDB — Vector Database from Scratch in C++

[GitHub](#)

- Built **header-only in-memory vector database** supporting vector insertion, deletion, and similarity search.
- Implemented **HNSW-based approximate nearest neighbor (ANN)** vector index enabling efficient high-dimensional embedding retrieval.
- Added disk persistence and CRUD operations enabling rapid experimentation with embedding pipelines.
- Achieved **~2 ms query latency for 512-dimensional embeddings**.

ResearchIT - Graph RAG Research Assistant

[Live Website](#)

- Built research assistant using **Graph-based Retrieval Augmented Generation architecture**.
- Combined **Pinecone vector search with Neo4j knowledge graph** to capture relationships between research methods, metrics, and claims.
- Enabled multi-paper querying across research documents using **LLM reasoning over structured graph data**.
- Deployed application using **FastAPI backend and Vercel frontend**.

PersonalOS — AI Productivity Assistant

- Built AI productivity assistant using **LangGraph orchestration with Google Gemini models**.
- Integrated Gmail, Google Calendar, and Notion APIs to automate scheduling and task management.
- Implemented **vector-based memory using Pinecone** for contextual recall and long-term knowledge storage.

EDUCATION

B.E. Computer Science and Design | CGPA - 8.31 | *July 2025*

K. K. Wagh Institute of Engineering Education and Research | *Nashik*

CERTIFICATIONS

- Full Stack Web Development (MERN Stack) - Udemy (Aug 2024)
- Python Programming Certification - Udemy (Oct 2024)
- AWS Cloud Technical Essentials - Coursera (Aug 2025)
- Machine Learning Specialization (Andrew Ng) - Coursera (Dec 2025)

ACHIEVEMENTS

- 1st Place MET Enthusia: Galaxy Morphology Classification & Redshift Estimation March 2025
- 1st Place IEEE Technovation: Galaxy Morphology Classification & Redshift Estimation Apr 2025